



StackMaster Tech

CLOUD · DATA · OBSERVABILITY · SRE

FIELD HANDBOOK · DAY 6 · CONCEPTS & THEORY

Security & Machine Learning Overview

A conceptual tour of the platform beyond core Splunk: **Splunk Cloud** topology and secure ingest, **Enterprise Security** as a SIEM, and the **Machine Learning Toolkit** for anomaly detection and forecasting. Theory-first — no lab required.

Splunk Cloud & Topology

Private Connectivity (AWS)

Enterprise Security · SIEM

Risk-Based Alerting

MITRE ATT&CK

MLTK · Forecast & Anomaly

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01 · Beyond Core Splunk — Cloud, SIEM & ML

Days 1–5 built fluency in core Splunk: searching, enriching, reporting, alerting and tuning. This overview looks outward to three big directions the platform grows in — **where Splunk runs** (Splunk Cloud), **a security application built on it** (Enterprise Security), and **analytics on top of it** (the Machine Learning Toolkit).

HOW TO READ THIS HANDBOOK

Splunk Cloud and Enterprise Security are covered

CONCEPTUALLY

— architecture, workflow and best practice — because they need a managed Cloud tenant or an ES licence to touch hands-on. The

MACHINE LEARNING TOOLKIT

section is different: MLTK is a free app you can run on any Splunk, so that part is

HANDS-ON

, with runnable labs on the same Buttercup Games dataset used all week.

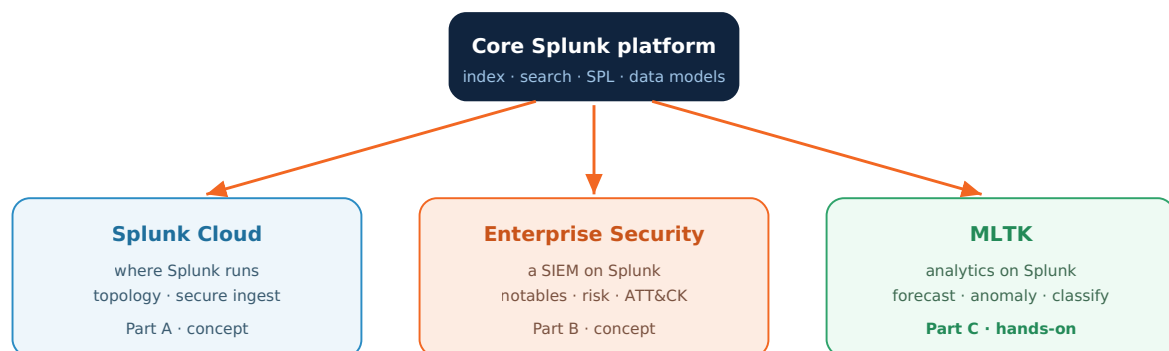


Fig 1.1 — Three directions the platform extends in — all resting on the same core you already know.

PART A

Splunk Cloud Overview

Splunk Cloud Platform is Splunk delivered as a managed SaaS service. The SPL and concepts are identical to Enterprise; what changes is **who runs the infrastructure, how data gets in securely, and where the responsibility boundary sits**. This part is conceptual — topology, ingestion strategy and secure connectivity.

02 · Splunk & Splunk Cloud — Features & Topology

In Splunk Enterprise you run every tier yourself: forwarders collect data, indexers parse and store it, and search heads run SPL against the indexers. Splunk Cloud keeps the same tiers but Splunk operates the indexer and search-head tiers for you; you manage data onboarding and the searches, dashboards and apps on top.

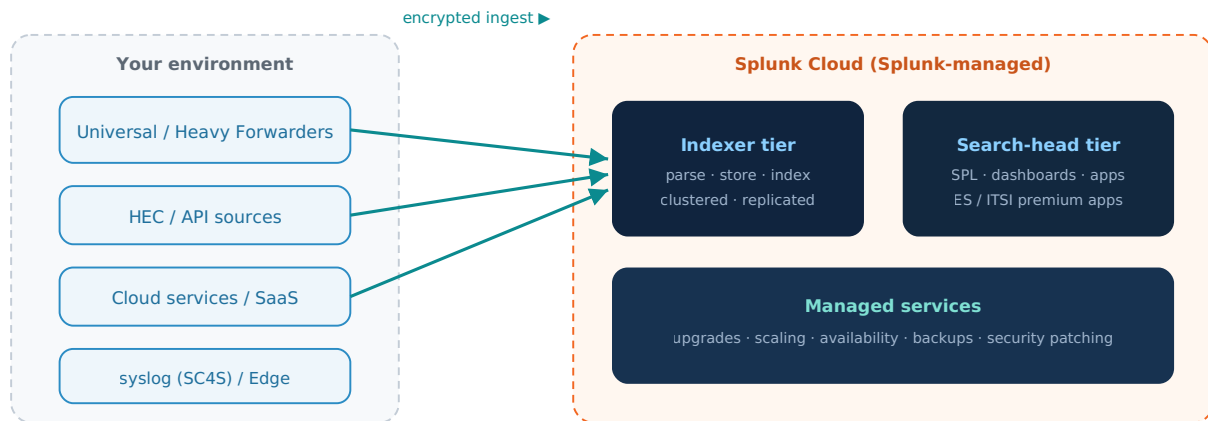


Fig 2.1 — Same tiers; the responsibility line moves. You own onboarding & content; Splunk operates indexers, search heads and platform ops.

Enterprise vs Cloud — the practical differences

Aspect	Splunk Enterprise	Splunk Cloud Platform
Who runs indexers/SH	You	Splunk (SaaS)
Upgrades & scaling	Your ops team	Handled by Splunk
Admin access	Full OS + .conf	Via UI, ACS API, and supported .conf; no OS shell
App install	Any app	Cloud-vetted apps (self-service or vetting)
Ingest	Direct to indexers	Forwarders/HEC over TLS to the tenant
SPL & data	Identical — searches, knowledge objects and dashboards port across	

SHARED-RESPONSIBILITY MINDSET

Cloud removes infrastructure toil but not data ownership: onboarding quality, source typing, field extractions, RBAC and search hygiene are still yours. Everything you learned Days 1–5 applies unchanged.

03 · Secure Splunk Deployment Best Practices

Whether self-managed or Cloud, a secure Splunk deployment follows the same principles: encrypt everything in transit, authenticate strongly, authorise narrowly, and keep an audit trail. Cloud handles the platform layer; these practices are what *you* own.

Domain	Best practice
Encryption in transit	TLS on all forwarder→indexer, HEC and web traffic; validate certificates, disable weak ciphers.
Authentication	SSO via SAML/OIDC; enforce MFA; avoid shared local accounts.
Authorisation (RBAC)	Least-privilege roles; scope index access with <code>srchIndexesAllowed</code> ; restrict risky capabilities (<code>delete_by_keyword</code> , <code>indexes_edit</code>).
Data segregation	Separate indexes per data sensitivity; role-to-index mapping; field masking for PII at index or search time.
Auditing	Monitor <code>index=_audit</code> ; alert on privilege changes, failed logins, config edits.
Network	Restrict management ports; private connectivity for ingest (Ch. 5); IP allow-lists on the tenant.
Secrets & onboarding	Rotate HEC tokens; scope tokens per source; avoid credentials in search strings or macros.

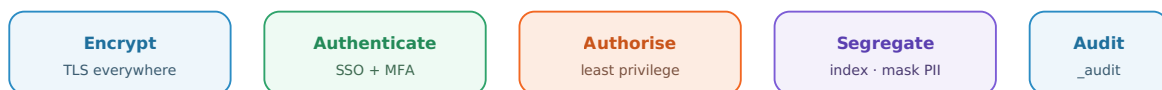


Fig 3.1 — The five pillars of a secure deployment, applied on top of the platform Splunk secures for you.

DEFENCE IN DEPTH

No single control is enough. A leaked HEC token is contained by TLS + IP allow-listing + per-source scoping + audit alerting — layer them so one failure isn't a breach.

04 • Splunk Cloud Data-Ingestion Strategies

Getting data into Splunk Cloud is the main thing you operate. Splunk offers several ingest paths; the right one depends on the source — an OS, an app, a SaaS API, a network device, or a cloud provider's telemetry.

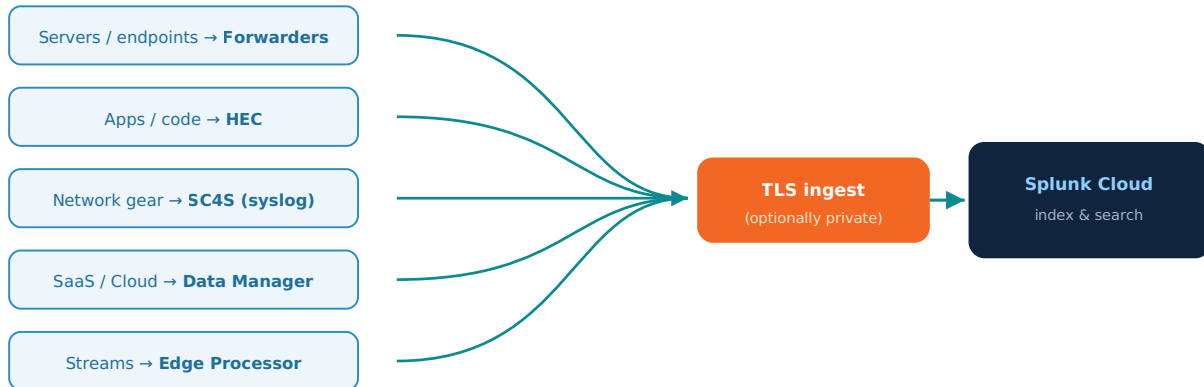


Fig 4.1 — Ingest paths converge on an encrypted (optionally private) channel into the Cloud tenant.

Path	Best for	Notes
Universal Forwarder	OS logs, files, metrics at scale	Lightweight agent; TLS + load-balanced to tenant.
HTTP Event Collector	Apps, containers, serverless, webhooks	Token-based JSON/raw over HTTPS.
SC4S (syslog)	Firewalls, routers, appliances	Splunk Connect for Syslog normalises then forwards.
Data Manager	AWS / Azure / GCP telemetry	Guided cloud onboarding with CIM mapping.
Edge Processor / Ingest Actions	Filter, mask, route <i>before</i> indexing	Cut noise & cost; redact PII in the pipeline.

ONBOARD WITH INTENT

Decide sourcetype, index, timestamp and CIM mapping *before* data flows. Clean onboarding is what makes Days 2–4 (fields, lookups, dashboards) and Enterprise Security work later — and Edge Processor/Ingest Actions let you drop or mask data to control cost and risk at the door.

05 • Secure Ingest — Private Connectivity with AWS

By default, forwarders reach Splunk Cloud over the public internet (encrypted with TLS). For regulated or sensitive data, **Splunk Cloud Private Connectivity** uses **AWS PrivateLink** to keep ingest traffic entirely on the AWS private network — it never traverses the public internet.



Traffic stays on the AWS backbone end-to-end — the public internet is never used.

Fig 5.1 — Private Connectivity: your VPC endpoint → AWS PrivateLink → Splunk's endpoint service, all on the private AWS network.

Property	Benefit
No public internet exposure	Ingest never leaves the AWS private network — smaller attack surface.
Private DNS & IPs	Forwarders resolve the tenant to a private endpoint in your subnet.
Regulatory alignment	Helps meet data-in-transit and network-isolation requirements.
Same forwarders	Existing UF/HEC config points at the private endpoint — minimal change.

STILL ENCRYPTED

Private Connectivity is about the *path* (private vs public), not a replacement for TLS. You keep transport encryption *and* gain network isolation — defence in depth for ingest.

WHEN TO USE IT

Reach for Private Connectivity when policy forbids sensitive telemetry crossing the public internet, or when you're already AWS-resident and want ingest to stay on the backbone. For most workloads, TLS over the internet is the simpler default.

PART B

Enterprise Security (SIEM)

Splunk Enterprise Security (ES) is a premium app that turns Splunk into a full **SIEM** — Security Information and Event Management. It ships correlation searches, risk scoring, threat-intelligence frameworks, ATT&CK mapping and an analyst workflow. This part is conceptual; the screenshots are *illustrative* of the ES experience.

06 · Introduction to Enterprise Security

A SIEM centralises security-relevant data, applies detections continuously, and gives analysts a workflow to investigate and respond. ES builds all of this on the Splunk you know: data is normalised to the **Common Information Model**, **correlation searches** generate **notable events**, and analysts triage them in **Incident Review**.

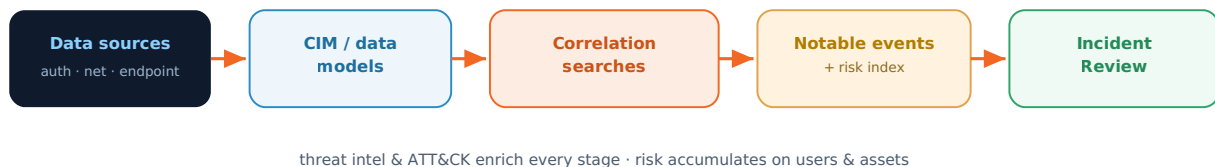


Fig 6.1 — The ES pipeline: normalise → detect → raise notables/risk → triage.

ES concept	What it is
Common Information Model	Shared field names/data models so detections work across any compliant source.
Correlation search	A scheduled search encoding a detection; fires notable events and/or risk.
Notable event	A triageable security finding with urgency, status and owner.
Risk index / RBA	Accumulates risk on entities (user, host, IP) from many small signals.
Threat intelligence	Frameworks to match events against known-bad indicators.
Security domains	Access, Endpoint, Network, Identity, Threat, Audit — how content is organised.

ES IS SPLUNK, SPECIALISED

Every ES detection is ultimately SPL over data models. The Days 1–5 skills — search, stats, lookups, data models, tstats — are exactly what you use to read, tune and author ES content.

07 · Real-Time Threat-Monitoring Dashboards

ES ships purpose-built dashboards. The **Security Posture** view is the at-a-glance health of the environment (key indicators, notable trends by domain); **Incident Review** is the working queue where analysts triage. Glass Tables visualise custom security KPIs against your architecture.

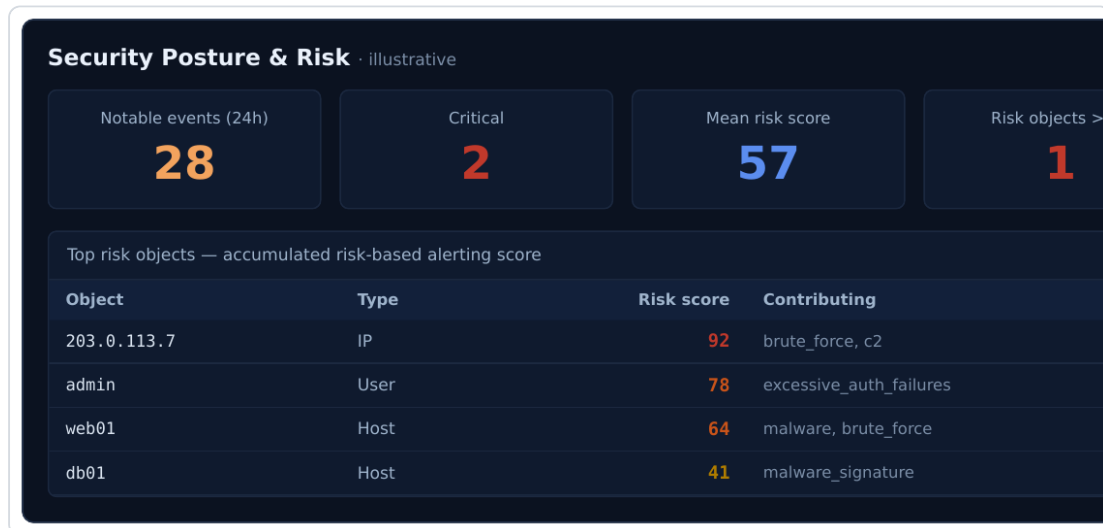


Fig 7.1 — Illustrative Security-Posture-style view: KPIs plus the top risk objects accumulating score.

Dashboard	Purpose
Security Posture	Executive/at-a-glance KPIs and notable-event trends by security domain.
Incident Review	The analyst queue: filter, assign, investigate, and update notable events.
Risk Analysis	Risk scores by object; which entities are heating up and why.
Glass Tables	Custom real-time security metrics laid over a topology you design.
Domain dashboards	Access / Endpoint / Network deep-dives for focused hunting.

BUILT ON WHAT YOU KNOW

These are Splunk dashboards over the ES data models — you can extend them with the Dashboard Studio skills from Day 4, adding panels, inputs and drilldowns to fit your SOC.

08 • Notable Events, Risk Scores & Triage

A **notable event** is the unit of work for an analyst. It carries an **urgency** (derived from severity × asset/identity priority), a **status** (New → In Progress → Resolved → Closed) and an **owner**. The triage workflow moves each notable from detection to disposition.

Incident Review — Enterprise Security · illustrative				
6 notable events · last 24h · 2 Critical · 2 High				
Time	Domain	Title	Urgency	Status
07-18 05:14	Access	Brute force attack — SSH source 203.0.113.7	Critical	New
07-18 05:22	Access	Excessive failed logins for user admin	High	In Progress
07-18 04:02	Network	Scan detected from 198.51.100.22	Medium	New
07-17 22:40	Endpoint	Malware signature match on host db01	High	In Progress
07-17 19:03	Threat	Connection to known C2 indicator	Critical	New
07-17 14:11	Identity	New privileged account created	Low	Closed

Fig 8.1 — Illustrative Incident Review: notable events with urgency, status and owner — the analyst queue. Note the brute-force finding from our week-long example, 203.0.113.7.



Fig 8.2 — The triage lifecycle: every notable moves left-to-right to a disposition, with audit at each step.

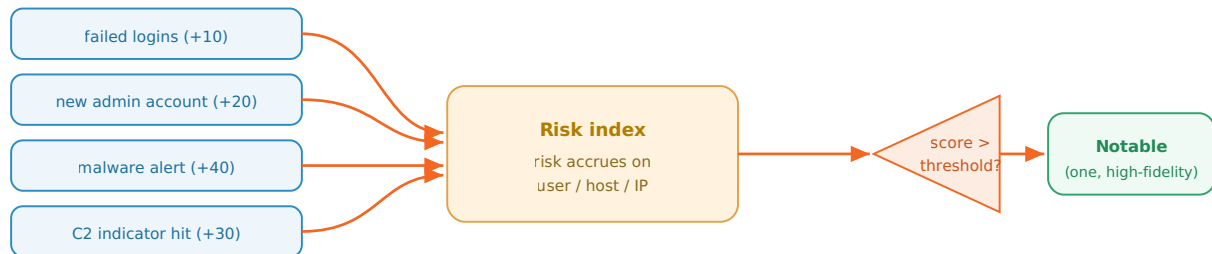
Risk scores add a second dimension. Rather than paging on every weak signal, ES can assign a small risk score to each and accumulate it on the entity. When a user or host crosses a threshold, *that* becomes a notable — fewer, higher-quality alerts. This is Risk-Based Alerting, next.

URGENCY, NOT JUST SEVERITY

A medium-severity detection on a crown-jewel server can outrank a high-severity one on a test box. ES computes urgency from severity *and* asset/identity priority — so triage order reflects business risk.

09 • Detection Rules & Risk-Based Alerting

Detections in ES are **correlation searches** — scheduled SPL that recognises a threat pattern and produces a notable, a risk contribution, or both. **Risk-Based Alerting (RBA)** is the modern approach: many searches write small risk scores to a risk index, and a single rule fires a notable only when accumulated risk crosses a threshold.



Many weak signals → accumulated risk → one meaningful alert (fewer false positives, more context).

Fig 9.1 — Risk-Based Alerting: aggregate small signals onto entities; alert on the total, not each event.

Analysing risk scores & alerts

When a risk notable fires, the analyst reads the **risk timeline** for that entity: which detections contributed, in what order, and mapped to which attack stages. A rising score that spans reconnaissance → access → exfiltration signals a real intrusion far more convincingly than any single alert.

Rule type	Detects
Threshold / static	Known-bad conditions (e.g. $\geq N$ failed logins in 5 min — our <code>203.0.113.7</code> case).
Anomaly / statistical	Deviations from baseline (rare process, unusual data volume) — often MLTK-assisted.
Threat-intel match	Events matching indicators from a feed (Ch. 10).
Risk-based (RBA)	Accumulated risk across many detections crossing a threshold.

WHY RBA WON

Traditional SIEMs drowned analysts in per-event alerts. RBA raises the signal-to-noise ratio: a single risk notable arrives with a built-in story — the contributing events and their ATT&CK context — so triage is faster and false positives fall.

10 • Threat Intelligence Feeds

Threat intelligence is the set of known-bad **indicators of compromise** (IoCs) — malicious IPs, domains, file hashes, URLs — that ES matches your data against. ES's **Threat Intelligence Framework** ingests feeds, normalises them into threat collections, and lets correlation searches flag any event touching an indicator.

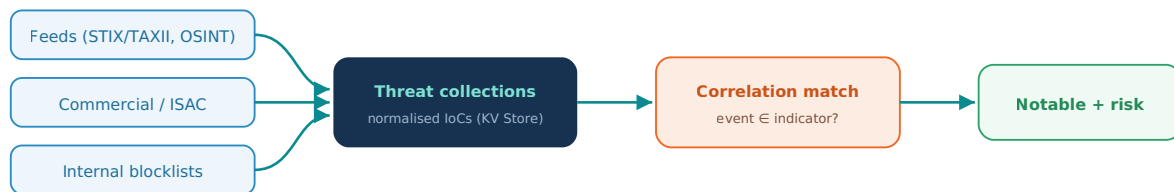


Fig 10.1 — Feeds normalise into threat collections; correlation searches match events and raise notables.

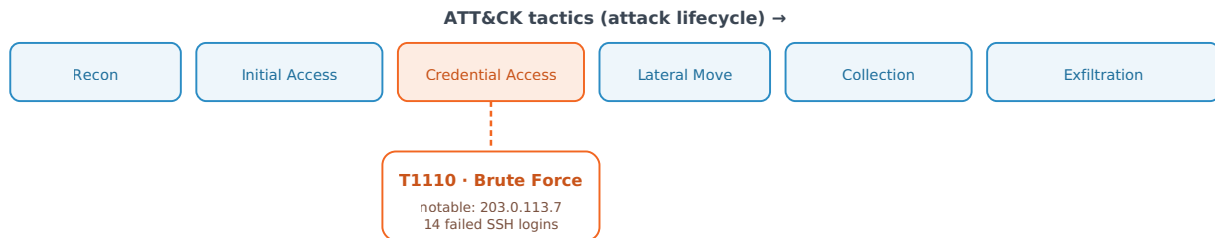
Indicator type	Example match
Network	Malicious IP / domain / URL seen in proxy or firewall logs.
File	Known-bad hash in endpoint telemetry.
Email / user	Phishing sender, compromised account indicator.
Certificate / registry	Malware infrastructure fingerprints.

FEEDS NEED HYGIENE

Stale or low-quality feeds cause false positives and alert fatigue. Age out old indicators, weight sources by confidence, and whitelist known-good to keep matches meaningful — the same lookup-hygiene discipline from Day 2, at security scale.

11 · Mapping Alerts to Attack Techniques (ATT&CK)

MITRE ATT&CK is a knowledge base of adversary **tactics** (the *why* — e.g. Initial Access, Persistence, Exfiltration) and **techniques** (the *how* — e.g. T1110 Brute Force). ES annotates detections with ATT&CK IDs so every notable says not just "something happened" but "which stage of an attack this represents".



Each detection is tagged with a technique; a risk story becomes a path across the tactic columns.

Fig 11.1 — Our brute-force example maps to ATT&CK technique T1110 under the Credential Access tactic.

Benefit of ATT&CK mapping	Why it matters
Common language	Analysts, detections and reports all speak in shared technique IDs.
Coverage gaps	An ATT&CK heat-map shows which techniques you can/can't detect.
Attack-chain context	A risk story that spans multiple tactics reads as a genuine intrusion.
Prioritisation	Focus detection engineering on the techniques most relevant to your threats.

FROM ALERT TO NARRATIVE

ATT&CK turns a pile of alerts into a story: "brute force (T1110) → valid accounts (T1078) → data staged (T1074) → exfiltration (T1041)." That narrative is what convinces responders — and it's built by tagging each detection with its technique.

PART C

Machine Learning Toolkit (MLTK)

The Machine Learning Toolkit is a **free** Splunk app that adds `fit` and `apply` — train a model with SPL, then score new data with it. Unlike Parts A and B, this part is **hands-on**: every lab runs on the same Buttercup Games dataset, and the screenshots are *real* results computed from it.

12 · MLTK Overview & the ML Workflow

MLTK wraps popular algorithms (scikit-learn plus Splunk's own) behind two SPL commands. `fit` trains a model from search results and saves it; `apply` runs a saved model against new results to predict, classify, forecast or flag. Around these sit guided **Assistants**, a **Showcase** of examples, and **Experiments** to manage models.

The two commands everything is built on

```
# train and save a model
... | fit <Algorithm> <target> FROM <features> into app:my_model
# score new data with the saved model
... | apply app:my_model
```

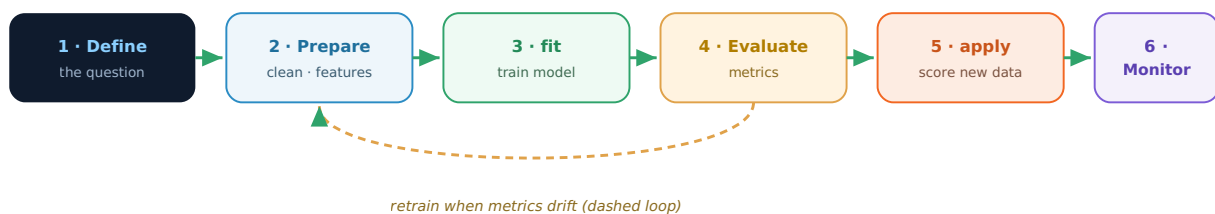


Fig 12.1 — The ML workflow. Most effort is stages 2 & 4 (prepare and evaluate), not the one-line `fit`.

Algorithm families in MLTK

Family	Example algorithms	Answers
Forecasting	ARIMA, Kalman filter, <code>predict</code> , StateSpaceForecast	"What comes next?" (Ch. 15)
Anomaly detection	DensityFunction, LocalOutlierFactor, OneClassSVM	"What's unusual?" (Ch. 14)
Regression	LinearRegression, RandomForestRegressor	"How much / how many?"
Classification	LogisticRegression, RandomForest, SVM	"Which category?" (Ch. 16)
Clustering	KMeans, DBSCAN, Spectral	"What groups exist?"

THREE WAYS TO USE MLTK ASSISTANTS

(guided point-and-click for each family), the

SHOWCASE

(worked examples on sample data), and raw

SPL

(`fit / apply` in any search or dashboard). The labs below use SPL so they drop straight into what you learned Days 1–5.

13 · Data Preparation & Transformation

Models learn from **features** — numeric, well-shaped columns. Preparation is 80% of real ML work: aggregate raw events to the right grain, engineer features, handle nulls, scale, and encode categories. In Splunk this is just SPL you already know (`stats`, `eval`, `fillnull`) plus a few MLTK preprocessors.

Task	SPL / MLTK	Why
Aggregate to grain	<code>timechart</code> / <code>stats BY</code>	One row per hour / session / entity
Engineer features	<code>eval</code> , <code>streamstats</code>	ratios, rolling stats, hour-of-day
Handle missing	<code>fillnull</code> , <code>fit Imputer</code>	models reject nulls
Scale numerics	<code>fit StandardScaler</code>	comparable ranges
Encode categories	<code>fit FieldSelector</code> / one-hot via <code>eval</code>	text → numeric

LAB 13 · BUILD A FEATURE TABLE FROM BUTTERCUP WEB TRAFFIC

Turn raw `access_combined` events into an hourly feature table ready for forecasting or anomaly detection.

```
index=web
| timechart span=1h count AS hits, dc(clientip) AS visitors, sum(price) AS revenue
| eval hour=strftime(_time,"%H"), rev_per_visitor=round(revenue/visitors,2)
| fillnull value=0
```

Each row is now one hour with clean numeric features (`hits`, `visitors`, `revenue`, `hour`, `rev_per_visitor`) — the input every model in this part expects.

GARBAGE IN, GARBAGE OUT

A model can't fix bad features. Leakage (a feature that secretly encodes the answer), wrong grain, and unscaled inputs cause more failures than algorithm choice. Spend your time here.

14 · Detecting Anomalies & Unusual Spikes

Anomaly detection answers "what's unusual?" MLTK's `DensityFunction` learns the normal distribution of a metric, then flags values in the improbable tails as outliers. It's the ML-grade successor to the fixed thresholds from Day 3 — it adapts to the data's own shape.

LAB 14 · FLAG ANOMALOUS LOGIN-FAILURE SPIKES

Learn the normal distribution of hourly failed logins on `index=security`, then flag the outlier hours — the same brute-force burst we've tracked all week, found statistically.

```
# 1) shape the metric: failed logins per hour
index=security vendor_action="Failed password"
| timechart span=1h count AS fails
# 2) fit a density model and flag the improbable tail (~0.5%)
| fit DensityFunction fails threshold=0.005 into app:ssh_fail_density
# 3) keep only the outliers
| where IsOutlier(fails)=1
```

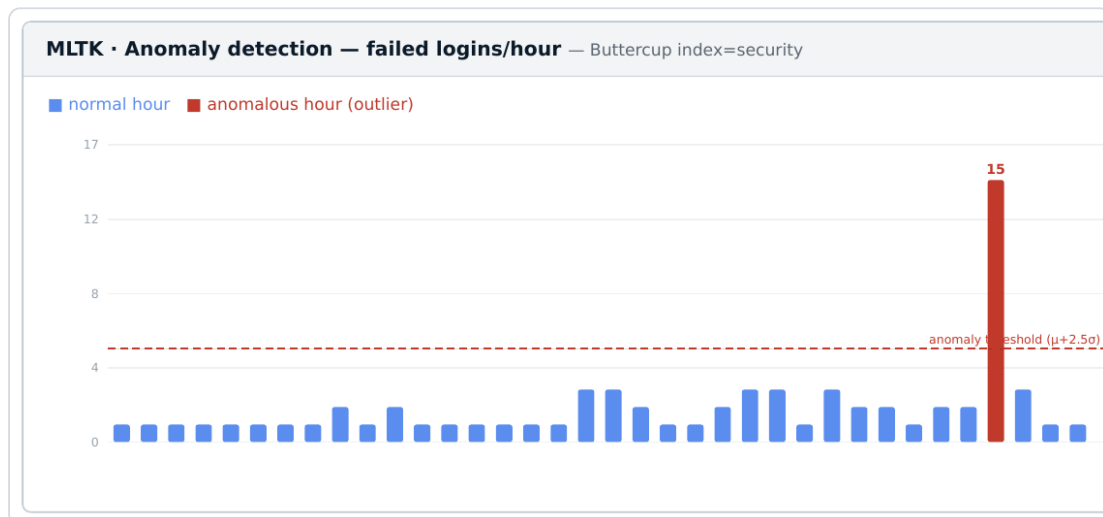


Fig 14.1 — Real Buttercup data: hourly failed logins with the $\mu+2.5\sigma$ threshold. Baseline $\approx 1.8/\text{hr}$; the 05:00 spike of 15 sits far above it.

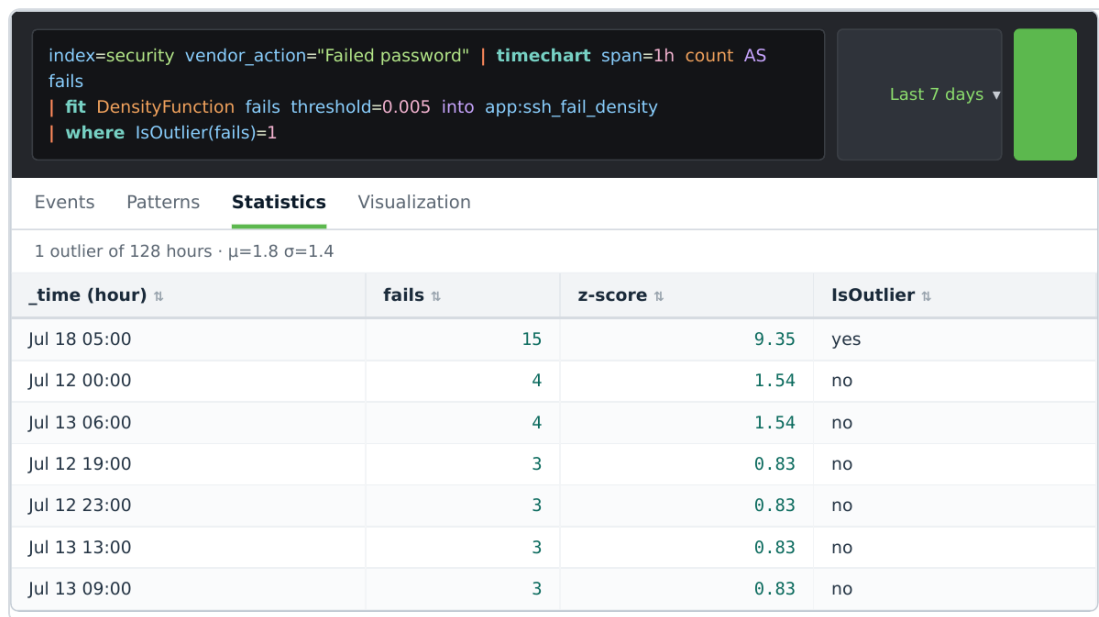


Fig 14.2 — Real `fit DensityFunction` output: 203.0.113.7's 05:00 hour is the lone outlier ($z \approx 9.35$) among 128 hours.

The model learned that the environment normally sees one or two failures an hour ($\mu \approx 1.8$, $\sigma \approx 1.4$). Fifteen in a single hour is ~ 9 standard deviations out — unambiguously anomalous, and flagged with no hand-tuned threshold.

Method	Use
<code>DensityFunction</code>	Single-metric outliers from a learned distribution (above).
<code>anomalydetection</code> / <code>outlier</code>	Quick built-in outlier flagging without a saved model.
<code>LocalOutlierFactor</code> / <code>OneClassSVM</code>	Multivariate anomalies across several features.
Time-series decomposition	Spikes relative to trend + seasonality (spot busy-hour vs true spike).

SUSPICIOUS PATTERNS, NOT JUST SPIKES

Feed multivariate features (hits, bytes, distinct URLs, error rate) to `LocalOutlierFactor` to catch behaviour that's odd *in combination* even when no single metric looks extreme — the essence of behavioural detection.

15 · Forecasting Trends & Predicting Values

Forecasting projects a time series forward — capacity planning, demand prediction, "is today tracking normal?" MLTK offers `predict` (state-space/Kalman) and algorithms like ARIMA; both return predicted values plus a confidence interval.

LAB 15 · FORECAST HOURLY WEB TRAFFIC

Model Buttercup's hourly hit volume and project the next day, then judge the forecast against held-out actuals.

```
# build the hourly series and forecast the next 24 points with 95% bounds
index=web
| timechart span=1h count AS hits
| predict hits future_timespan=24 holdback=24 upper95=u lower95=l
```

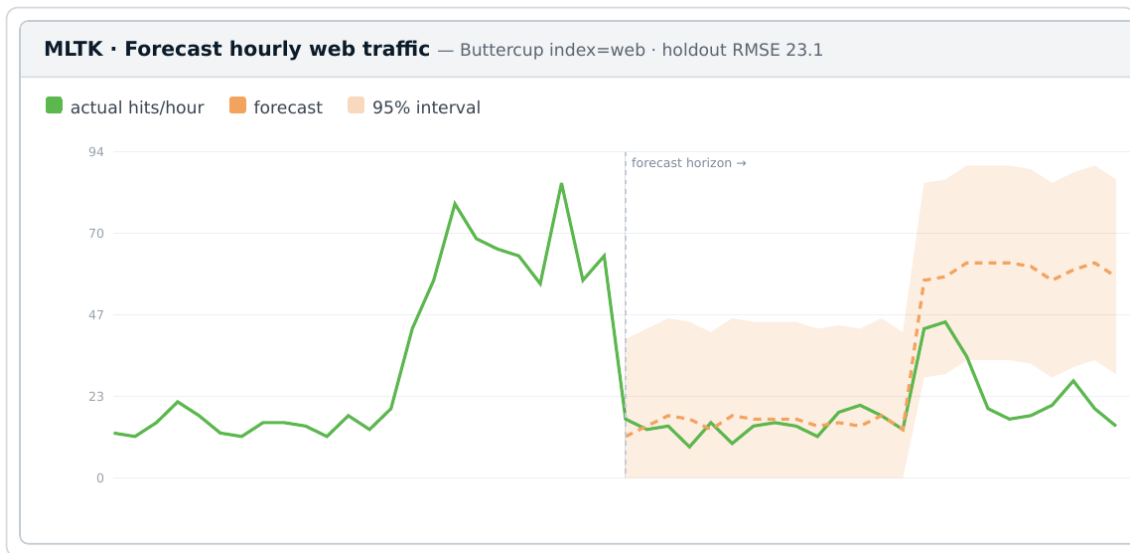


Fig 15.1 — Real Buttercup traffic: the model captures the daily shape and projects the next 24 h with a 95% band (holdout RMSE $\approx$ 23.1 hits).

The `holdback=24` option hides the last 24 hours from training so the forecast can be scored against them — the honest way to know if a forecast is any good *before* trusting it live. Here the model tracks the diurnal rhythm; wider bands mark hours of greater uncertainty.

Option	Effect
<code>future_timespan</code>	How many points to project forward.
<code>holdback</code>	Hide the last N points from training for validation.
<code>upper95</code> / <code>lower95</code>	Confidence-interval fields — the uncertainty around each prediction.
<code>period</code>	Seasonality length (e.g. 24 for hourly-with-daily-cycle).

FORECASTS ASSUME THE FUTURE RESEMBLES THE PAST

A model trained on a quiet week won't foresee a launch spike or an outage. Pair forecasts with the anomaly detection of Ch. 14: predict the expected band, then alert when reality breaks out of it.

16 · Model Training, Evaluation & Metrics

Training is one line; **evaluation** is what separates a useful model from a dangerous one. You split data into train/test, fit on train, score on the unseen test, and read the right metric for the task — regression/forecast errors differ from classification metrics.

Train / test discipline

```
# fit on a partition, apply to the held-out rest
... | sample partitions=10 seed=42
| eval role=if(partition_number<7,"train","test")
# fit on role=train ... then apply to role=test and score
```

Which metric for which task

Metric	Task	Reads as
RMSE / MAE	Regression / forecast	Average error in the target's units — lower is better. <i>Our forecast: RMSE $\approx$ 23.1 hits.</i>
Accuracy	Classification	% correct overall — misleading on imbalanced data.
Precision	Classification	Of predicted positives, how many were right — $TP/(TP+FP)$.
Recall	Classification	Of actual positives, how many were caught — $TP/(TP+FN)$.
F1	Classification	Harmonic mean of precision & recall — one balanced number.

The confusion matrix — where classification metrics come from

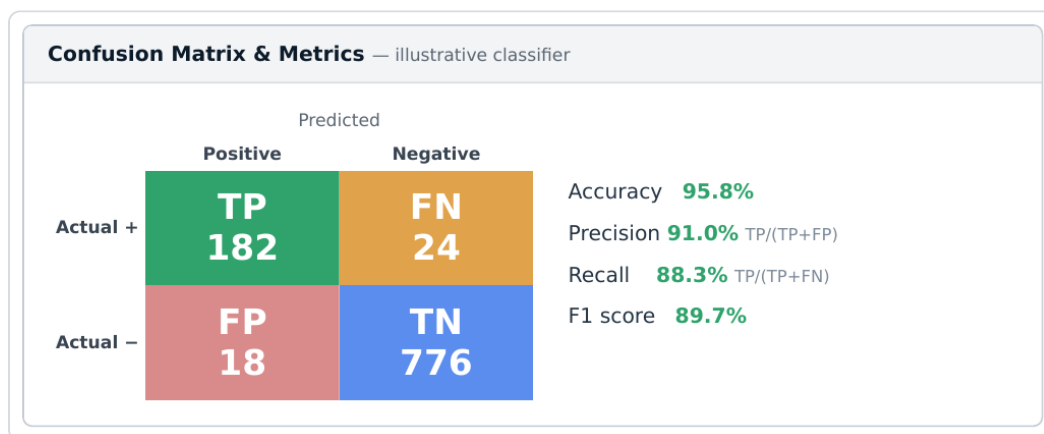


Fig 16.1 — Illustrative classifier: every classification metric is derived from the four cells (TP / FP / FN / TN).

LAB 16 · SCORE A CLASSIFIER — AND READ A RED FLAG

Train a session-level model on Buttercup to predict which sessions end in a purchase (features: views, add-to-carts, removes, distinct categories).

```
index=web | stats count(eval(action="view")) AS views,
              count(eval(action="addtocart")) AS addtocarts,
              count(eval(action="purchase")) AS buys BY JSESSIONID
| eval label=if(buys>0,1,0)
```

```
| fit LogisticRegression label FROM views addtocarts into app:purchase_clf  
| apply app:purchase_clf | eval correct=if(label=predicted(label),1,0)
```

A REAL RESULT WORTH UNDERSTANDING

On our data this model scores ~100% accuracy — because a session's add-to-cart and purchase are structurally linked, the label is almost perfectly implied by the features. In practice a "perfect" score is a

WARNING SIGN

, not a trophy: suspect *target leakage* (a feature that encodes the answer). The fix is to evaluate on truly unseen data and remove features that wouldn't be known at prediction time. This is exactly why evaluation discipline matters more than the one-line `fit`.

MATCH METRIC TO COST

A fraud/threat detector usually favours

RECALL

(don't miss attacks) even at some precision cost; a marketing model may favour

PRECISION

(don't waste spend). On imbalanced data, ignore accuracy and read precision/recall — 99% accuracy is worthless if it never catches the 1% that matters.

17 · Appendix A — Where Each Capability Fits

A quick map from a goal to the right part of the platform.

You want to...	Reach for	Covered in
Run Splunk without managing servers	Splunk Cloud Platform	Ch. 2
Keep ingest off the public internet	Private Connectivity (AWS PrivateLink)	Ch. 5
Onboard OS / app / cloud data securely	Forwarders · HEC · Data Manager · Edge	Ch. 4
Run a SOC with triage & risk	Enterprise Security	Ch. 6–9
Match events to known-bad indicators	Threat Intelligence Framework	Ch. 10
Speak the language of attacks	MITRE ATT&CK mapping	Ch. 11
Find the unusual automatically	MLTK · DensityFunction / outliers	Ch. 14
Predict what comes next	MLTK · predict / ARIMA	Ch. 15
Classify or score with a model	MLTK · fit / apply + evaluation	Ch. 16

THEY COMPOSE

The strongest deployments combine all three: Cloud runs the platform, ES provides the security workflow, and MLTK powers the anomaly/behaviour detections that feed ES risk — one pipeline from raw event to prioritised, explainable alert.

18 · Appendix B — Concept Glossary & Cheat Sheet

CLOUD & INGEST

Term	Meaning
SaaS tiers	Splunk runs indexers + search heads
HEC	token-based HTTP ingest
SC4S	syslog onboarding
Data Manager	guided cloud onboarding
PrivateLink	ingest stays on AWS backbone

ENTERPRISE SECURITY

Term	Meaning
CIM	shared field/model schema
Correlation search	a detection in SPL
Notable event	triageable finding
RBA	accumulate risk → alert
ATT&CK	tactic/technique taxonomy

MLTK — THE TWO COMMANDS

```
| fit Algo target FROM f1 f2 into app:m
| apply app:m
```

MLTK — LABS ON BUTTERCUP

```
# anomaly
| timechart span=1h count AS fails
| fit DensityFunction fails threshold=0.005
# forecast
| timechart span=1h count AS hits
| predict hits future_timespan=24 holdback=24
# classify + evaluate
| fit LogisticRegression label FROM views
addtocarts
```

METRICS AT A GLANCE

Metric	For
RMSE / MAE	forecast / regression
Precision	TP/(TP+FP)
Recall	TP/(TP+FN)
F1	balance of both

END OF THE OVERVIEW

You now have a conceptual map of Splunk Cloud (topology, secure ingest, Private Connectivity), Enterprise Security (notables, risk-based alerting, threat intel, ATT&CK), and hands-on fluency with the MLTK workflow — anomaly detection, forecasting, and honest model evaluation — all demonstrated on the Buttercup Games dataset from [Buttercup_Games_Dataset.zip](#).